

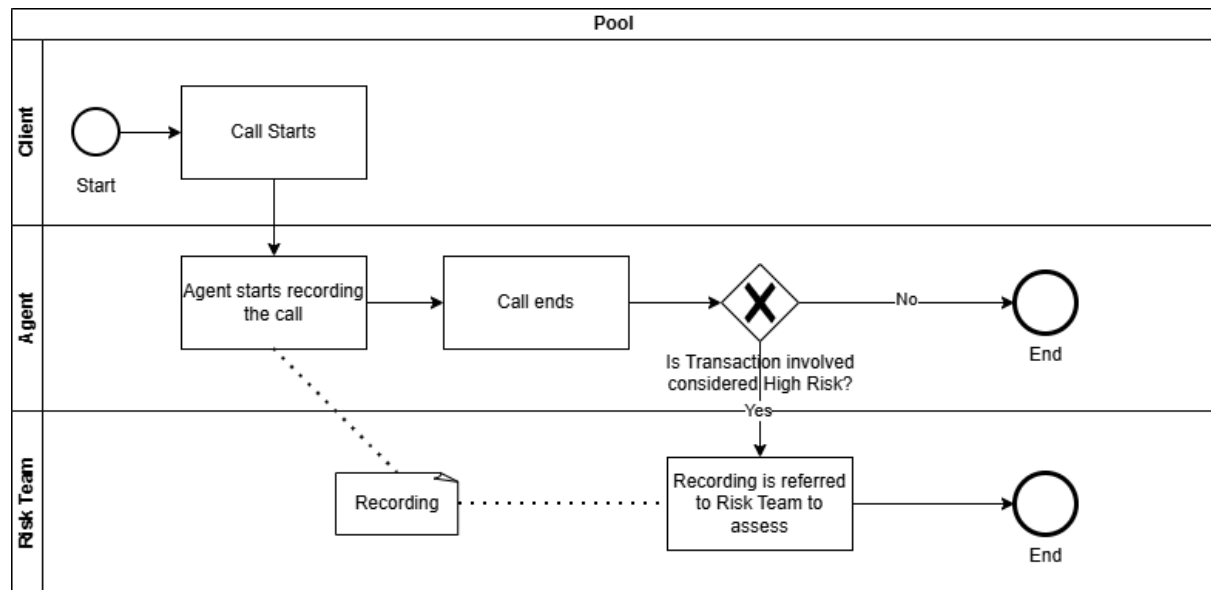
Adapted from FYP Project

Executive Summary:

Ensure confidentiality of bank-client calls while ensuring that it is usable for training transcription model

Background and Context:

“Data is the new oil” is an adage that collected data is valuable for a business’s continued growth and success. However, what is omitted is how the data is obtained, as well as the security involved.



In the banking industry, calls between clients and bank officers are recorded. If the call is high risk, the recording of the call is transcribed and then referred to the compliance team to assess before the transaction is approved. This is to ensure that proper disclosures are given to the client, and the bank does its due diligence before approval. Otherwise, the **bank-client relationship is strictly confidential – no one else should listen in to learn of clients’ sensitive data** (E.g. Bank Account, funds involved)

Additionally, as part of internal policies, employees are given security ratings based on their interaction with sensitive data. E.g. Agents -> Red (Confidential), Compliance Team -> Red (Confidential), Data Scientists -> Green (Publicly Available)

Emerging Use Case: Model Training

At the same time, the bank wishes to improve its internal transcription model. While accuracy is high for English speaking conversations, the transcription accuracy is significantly worse in the Asian context due to accents and regional dialect.

The current aim is to ensure that transcripts and recordings can allow data scientists and model to be **able to leverage the transcripts to improve accuracy while ensuring confidentiality**

Focus of our project: Ensure Confidentiality and Integrity of Data

Techniques to consider:

- Data Anonymisation: Anonymise the data when certain users are accessing it
 - Compliance team, Model: No change
 - Data scientist: Anonymised, changed certain parts that are identified to be sensitive data
- Attribute based encryption (Data at rest): Encrypt the data so that different users can have different level of access depending on their security clearance
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